

$$\frac{\partial C}{\partial w_{jk}} = z(a_j - y_j) \cdot \sigma'(z_j) \cdot a_k$$

For Bias:

$$\frac{\partial z}{\partial b} = 1 \rightarrow \frac{\partial C}{\partial b_j} = z(a_j - y_j) \cdot \sigma'(z_j)$$

Given a weight one layer back, w_{kr}

$$\frac{\partial C}{\partial w_{kr}} = \frac{\partial C}{\partial a_k} \cdot \frac{\partial a_k}{\partial z_k} \cdot \frac{\partial z_k}{\partial w_{kr}}$$



$$\frac{\partial C}{\partial a_k} = \frac{\partial C}{\partial a_j} \cdot \frac{\partial a_j}{\partial z_j} \cdot \frac{\partial z_j}{\partial a_k} = z(a_j - y_j) \cdot \sigma'(z_j) \cdot w_{jk}$$

(Same formula as before)

For Bias:

$$\frac{\partial C}{\partial b_k} = \frac{\partial C}{\partial a_k} \cdot \frac{\partial a_k}{\partial z_k}$$

- Parameters update throughout entire model
 - Possible via the chain rule of calculus
- Propagating partial derivatives backwards through neuron layers